CPRE 2300 HW 2
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A) The hash type is SHA-512. The 2nd field in the shadow file data contains the hashed password which is then divided into sections seperated by dollar signs. The first dollar sign contains the id. That id is used to find the hashing algorithm and in this case it was 6 which is SHA-512.

B) Similarly to the above, the 2nd section (after the 6) stores the salt. So in this case the salt becomes xgLS35S6.

C) I did not use OpenSSL tool to find the password initially as it mentioned in the hints for this homework that I can use Python and the crypt module for cracking the actual password. It's output:

```
Repos/2300-hashing △ ● python password_cracker.py
Started Searching
Password is: P@ssw0rd
Hash: $6$xgLS35S6$2UjEq.dUhICPw9zgDVJXcQYQp/9ilLPQt/8Zgu0uwngI5mVvB1eKQG9SnVLjmOOfkB4Jjb5VSAXGXjY4Cf5k90
Time taken: 0:00:00.568794
```

I then ran the OpenSSL tool to prove that the cracked password I generated produces the exact same hash:

```
Repos/2300-hashing △ ● openssl passwd -6 -salt xgLS35S6 "P@ssw0rd"
$6$xgLS35S6$2UjEq.dUhICPw9zgDVJXcQYQp/9ilLPQt/8Zgu0uwngI5mVvB1eKQG9SnVLjmOOfkB4Jjb5VSAXGXjY4Cf5k90
```

D) I have attached the script in the ZIP.

E) The password is - "P@ssw0rd"